



MAIS: Exploring human-AI interaction in fair and transparent recruitment with a multi-agent LLM-based system

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HIGHLIGHTS

- Introduces MAIS, an AI-powered interview platform to address scalability.
- Employs a modular multi-agent architecture with monitoring and safeguards.
- Conducts semi-structured interviews supporting multi-channel, multilingual input.
- Evaluated with 22 participants, showing high usability and positive experience.
- Embeds fairness, accountability, and robustness into AI-driven recruitment.

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ABSTRACT

The increasing volume of job applications received by organizations presents significant logistical and ethical challenges for recruitment processes. Traditional methods of screening and interviewing are often inefficient, inconsistent, and susceptible to bias, especially at scale. In this work, we introduce the Multi-Agent Interviewing System (MAIS), an AI-powered interview platform designed to support scalable and fair recruitment while preserving the human experience at the center of the process. MAIS combines multilingual, multi-channel communication with real-time monitoring and safeguards against misuse, enabling candidates to interact naturally with an AI interviewer through speech or text. We conducted a two-part user study involving 22 participants to explore both the usability of MAIS and the human perceptions that arise during interaction with an AI interviewer. Participants reported high usability and engagement, while qualitative feedback revealed themes of trust, transparency, and perceived fairness. These findings highlight not only the potential of MAIS to reduce recruiter workload and enhance consistency but also provide new insights into human-AI interaction dynamics in evaluative, high-stakes contexts. This work contributes to the design of safer, more transparent, and human-centered generative AI applications in recruitment and beyond.

1. Introduction

One pressing issue faced by many companies today is the need to efficiently manage a high volume of job applications across multiple roles. Even mid-sized organizations often receive hundreds of applications per vacancy, while large enterprises, such as Amazon, can attract

over 380,000 applications after a single recruiting event.¹ In October 2018, Amazon reported 850,000 applications for hourly roles in just one month, doubling its previous record.² Similarly, during their September 2021 Career Day, over 1 million candidates applied in 24 hours.³

Screening, interviewing, and evaluating such large candidate pools is both labor-intensive and time-consuming. According to industry data,

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¹ <https://press.aboutamazon.com/2020/9/amazon-career-day-2020-more-than-300-000-people-attended-the-event-and-over-380-000-applied-for-a-job-at-amazon>.

² <https://www.geekwire.com/2019/amazon-got-850k-job-applications-announcing-15-minimum-wage>.

³ <https://press.aboutamazon.com/2021/9/amazon-career-day-2021-more-than-1-million-people-apply-for-jobs-at-amazon>.

recruiters spend on average 7.4 s reviewing each resume,⁴ and only 2% of applications result in an interview invite, highlighting how challenging it is to manage these volumes manually.⁵ Additionally, the typical interview session lasts 30–60min, further compounding the time burden for HR teams.

The challenge is not only logistical but also qualitative: maintaining consistency, fairness, and responsiveness throughout the hiring pipeline becomes increasingly difficult at scale. Research shows that unstructured interview processes can introduce significant bias (Kausel et al., 2016), and compounded decision pipelines may perpetuate unfairness, even when individual components are designed to be fair (Bower et al., 2017). This underscores that as application volumes grow, ensuring equitable and transparent hiring becomes both practically and ethically complex.

In response to these challenges, the rapid advancement of large language models (LLMs) has opened new possibilities for automating and augmenting human resource processes (Sun, 2024). These models have unlocked a new era of interactive, intelligent systems capable of performing tasks once thought to require human intuition and judgment. From automated content generation to real-time conversation, LLM-powered agents are increasingly embedded into sensitive, high-stakes domains, such as recruitment (Alonso et al., 2024), education (Xing et al., 2025), and healthcare (Busch et al., 2025). However, the more capable these systems become, the greater the number of associated risks that emerge (Yao et al., 2024; Schillaci, 2024). Recent research highlights substantial concerns related to bias (Wei et al., 2025), misinformation (Chen and Shu, 2024), and social manipulation (Zhang et al., 2024), particularly when generative AI is used in decision-making processes that directly impact individuals.

These risks are particularly significant in the context of AI-powered recruitment. Automated interview systems must not only interpret complex human signals—linguistic, acoustic, and contextual—but also ensure fair, unbiased, and transparent evaluation processes (Booth et al., 2021). If implemented carelessly, generative models may reinforce stereotypes, be vulnerable to adversarial input, or amplify systemic inequities, ultimately compromising trust in the hiring process and potentially exposing organizations to reputational and legal risks (Hacker et al., 2025).

While automation offers scalability and efficiency, recruitment remains an inherently human-centered process, in which perceptions of empathy, fairness, and trust strongly influence acceptance of AI systems. Understanding how individuals experience interactions with AI interviewers is therefore crucial for designing solutions that complement rather than replace human judgment.

In this work, we aim not only to demonstrate a scalable technical solution but also to explore how humans engage with and perceive AI-mediated interviews, identifying key factors that shape usability, trust, and ethical alignment.

Therefore, this paper introduces a **Multi-Agent Interviewing System (MAIS)**, an AI-powered interview platform specifically designed to address both the scalability demands and ethical concerns of AI-driven recruitment. Rather than relying on black-box models, MAIS adopts a modular architecture that supports real-time monitoring, multilingual and multi-channel communication, enabling candidates to either speak or type during the interview, with speech automatically transcribed to text via ASR before processing, and proactive defense mechanisms against prompt injection and misuse. While the current analytic layer operates on textual data, this multi-channel design allows for flexible, accessible, and natural human-AI interaction during interviews.

By embedding ethical and safety considerations directly into the design, MAIS contributes to the growing body of work seeking to align generative AI systems with principles of fairness, robustness, and accountability.

More in detail, the contributions we bring in this paper are the following:

- We present an agentic architecture for conducting semi-structured job interviews, utilizing three specialized agents to conduct, monitor, and evaluate candidate responses in real-time.
- We conducted a two-part user evaluation involving 22 participants from diverse academic backgrounds, combining quantitative usability metrics such as System Usability Scale (SUS) with qualitative feedback to uncover how participants experience and interpret AI-driven interviews.
- We show several scenarios in which our system can be applied for large-scale screening and individual candidate assessment, thus reducing both the time required for screenings and the workload of recruiters.
- We indicate which ethical concerns arise in AI-driven recruitment, such as bias, transparency, and security, and how MAIS addresses them.
- We contribute to research on making generative AI safer and more transparent for important human-centered applications.

The remainder of this manuscript is structured as follows. [Section 2](#) reviews the background and related work in the field. [Section 3](#) presents the proposed architecture of MAIS, detailing its core components and design rationale. [Section 4](#) discusses the user evaluation conducted to assess the system's performance and usability. In [Section 5](#), we illustrate potential usage scenarios that highlight the flexibility and applicability of MAIS in real-world recruitment settings. [Section 6](#) and [Section 7](#) discuss the regulatory and ethical aspects related to the deployment and use of AI virtual interviewers, respectively. [Section 8](#) provides a theory-driven discussion that situates the system within the Human-AI Interaction literature, focusing on trust, social presence, and procedural justice in AI-mediated recruitment. Finally, [Section 9](#) concludes the paper with a summary of our contributions and highlights directions for future work.

2. Related work

In this section, we review related work across three intersecting domains: i) automated recruitment systems, ii) the application of LLMs in recruitment and candidate interaction, and iii) the ethical, legal, and social implications of deploying generative AI in hiring.

2.1. Automated recruitment systems

Over the past two decades, a wide range of efforts have aimed to automate various stages of the recruitment pipeline, from resume screening to candidate evaluation. Early systems predominantly relied on rule-based filtering and keyword-matching mechanisms, often embedded within Applicant Tracking Systems (ATS) (Maree et al., 2019). While effective at quickly reducing large applicant pools, these tools have been criticized for their lack of contextual understanding and for favoring candidates who optimize their resumes with specific keywords rather than demonstrating genuine suitability for the role.⁶

In response to these limitations, more recent approaches have leveraged machine learning to develop ranking systems that analyze natural language features extracted from resumes, cover letters, and job descriptions (Sinha et al., 2021). Some systems extend beyond document analysis, incorporating behavioral assessments, gamified psychometrics, and

⁴ <https://hrdailyadvisor.com/2018/11/15/eye-tracking-recruiters-average-7-4-seconds-reviewing-a-resume/>.

⁵ <https://www.myjobmag.com/blog/762/40-recruitment-statistics-you-need-to-get-hired-in-2019>.

⁶ <https://janzz.technology/are-you-disappointed-with-the-keyword-based-application-tracking-systems-ats-here-is-what-you-can-do/>.

structured digital interviews to more holistically evaluate a candidate's potential (Agrawal et al., 2020; Landers and Sanchez, 2022).

A particularly prominent development in this space is the emergence of asynchronous video interview (AVI) platforms, such as HireVue⁷ and myInterview.⁸ These systems use AI to analyze pre-recorded candidate responses, focusing on features such as facial expressions, vocal tone, and eye movement (Suen et al., 2020). While these modalities promise deeper behavioral insights, they have raised concerns regarding their scientific validity and the potential for discriminatory outcomes, particularly when deployed as replacements for trained human assessors (Díaz-Rodríguez et al., 2023).

Supporting this, Suen et al. (2024) demonstrates that AI can detect whether candidates are “selling” themselves or lying based on facial expressions and head movements more accurately than human interviewers. This highlights both the potential of AI to augment human judgment and the necessity of rigorous evaluation to ensure fair and ethical use. Recent research has started to shift from purely performance-oriented metrics toward psychologically informed frameworks for candidate training and evaluation. For example, Rao et al. (2024) introduced InterviewApp, a digital interview training platform that emphasizes supporting candidates' autonomy and competence during AI-mediated interview practice. Their findings highlight that perceived autonomy mediates the effectiveness of AI-based training tools, suggesting that interview systems should be designed not only for efficiency but also for motivational support and user well-being.

To address the limitations of keyword-based and surface-level analysis, embedding-based models such as Resume2Vec (Bevara et al., 2025) have been proposed. These systems map resumes and job descriptions into a shared vector space, allowing for more nuanced semantic matching. By capturing contextual and role-specific information, this approach significantly improves upon traditional ATS by moving beyond simple keyword overlap and enabling a deeper understanding of candidate relevance.

2.2. LLMs in recruitment and candidate interaction

The integration of LLMs into human resource (HR) systems is a relatively recent development that is rapidly gaining momentum. LLMs offer a powerful new interface for interpreting open-ended text, generating relevant follow-up questions, and simulating realistic interview dialogues (Hasegawa et al., 2025). Recent research has explored their utility in several key areas, including the automated creation of job descriptions, the tailoring of interview questions to role-specific competencies (V et al., 2024), and the evaluation of candidate responses along dimensions such as clarity, persuasiveness, and empathy (Xie et al., 2024).

Beyond candidate interaction, LLMs show promise in augmenting recruiter workflows. They can be used to generate dynamic summaries of candidate profiles, flag discrepancies between resume content and verbal responses, and assist in synthesizing large volumes of applicant data. Some emerging platforms have gone further by deploying LLM-powered avatars or conversational agents to conduct mock interviews for candidate preparation or to act as automated screeners in live interview settings (Li et al., 2023; Ashrafi et al., 2024; Daryanto et al., 2025).

Complementing these applications, agent-based frameworks have been proposed to enhance interpretability and decision support in the recruitment pipeline (Lo et al., 2025). These systems extract and summarize key information from resumes, provide candidate evaluations, and assign scores that closely align with those of human recruiters. In more advanced setups, LLMs have been used to simulate both the interviewer and the candidate based on the resume and job description (Sun et al., 2024), enabling the extraction of behavioral and motivational cues and

offering a richer basis for candidate-job matching beyond surface-level textual comparison.

Despite these advancements, most LLM-based recruitment systems continue to operate as black boxes, offering limited transparency into their decision-making processes. This lack of interpretability poses a significant challenge in high-stakes domains like hiring, where even perceived unfairness can undermine trust and expose organizations to reputational or legal risks (Yao et al., 2024; Hacker et al., 2025). Furthermore, many current tools still fall short in implementing real-time safety features such as bias detection, prompt injection mitigation, and defenses against adversarial candidate behaviour (Bullwinkel et al., 2025).

At the organizational level, Hina et al. (2025) found that successful AI assimilation in hiring depends on balancing technical capabilities with effective human-AI collaboration. Their study highlighted that enablers such as AI competency and recruitment agility promote adoption, whereas barriers like system opacity and low empathy hinder it. These findings underscore the need for transparent and collaborative frameworks, such as MAIS, to support effective integration.

2.3. Ethical and regulatory considerations in AI-driven hiring

The adoption of AI in recruitment raises profound ethical questions, particularly around bias, discrimination, and due process. As documented by several reviews (Floridi et al., 2018; Liu et al., 2025), automated decision-making systems often encode historical patterns of inequality present in training data or organizational culture. Even well-intentioned models trained on large corpora can reinforce gendered, racial, or ableist stereotypes unless specifically constrained (Binns et al., 2018).

Recent work by Xu et al. (2025) expands this discussion by examining how fairness is perceived in AI-mediated interviews. Through experimental studies, they demonstrate that candidates' perceptions of fairness depend on three key dimensions: AI literacy, the quality of system explanations, and the presence of interpersonal interaction during the interview. Their results suggest that even technically unbiased systems can be viewed as unfair if users lack understanding or feel disconnected from the process—underscoring the importance of transparency, explainability, and human-centered interaction design in AI hiring systems.

In parallel, Canagasuriam and Lukacik (2025) examined the use of ChatGPT in asynchronous video interviews as a form of AI-assisted cheating. They found that while AI-generated responses improved content quality, candidates who used ChatGPT were perceived as less honest and rated the process as less procedurally fair, highlighting emerging ethical risks in AI-mediated recruitment.

Building on this line of research, Suen and Hung (2024) investigated how different AI interface designs in asynchronous video interviews influence candidates' honest and deceptive impression management behaviours. Their findings reveal that the design and transparency of AI interfaces can actively shape how candidates present themselves, encouraging either authenticity or strategic self-enhancement. These results suggest that AI systems in hiring are not neutral observers, but active participants shaping behavioral outcomes.

Complementarily, Xiong and Kim (2025) examined how disclosure and message framing around AI use in hiring affect applicant attitudes and intentions. Across three experiments, they found that simply informing candidates that AI is used in recruitment tends to reduce organizational attractiveness unless the message is framed to emphasize fairness, transparency, and candidate benefits. Their findings highlight that communication design, not just algorithmic design, is critical to ensuring candidate trust and engagement in AI-mediated hiring processes.

In the context of hiring, bias amplification can have direct legal consequences. Regulatory bodies in the U.S. and EU have begun scrutinizing

⁷ <https://www.hirevue.com/>.

⁸ <https://www.myinterview.com/>.

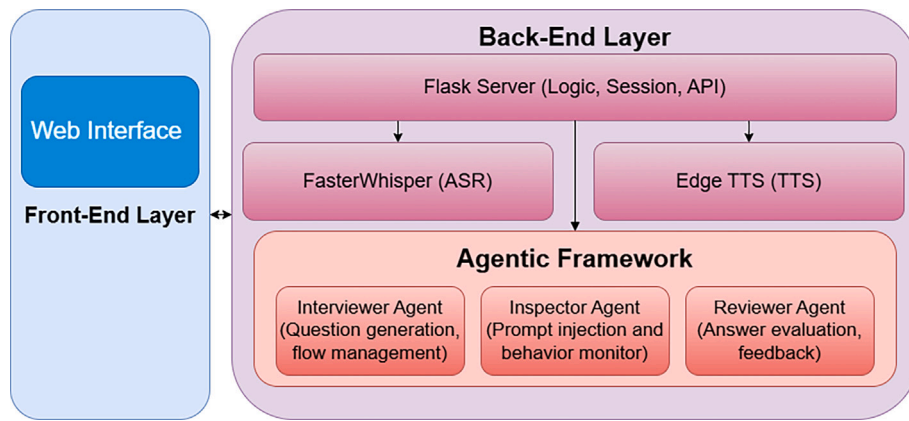


Fig. 1. Architecture of MAIS.

AI hiring tools under laws such as Title VII of the Civil Rights Act⁹ or the EU AI Act, which classify employment-related systems as “high-risk”.¹⁰ These frameworks emphasize the need for explainability, auditability, and human oversight in automated decisions. Some jurisdictions, such as New York City, have already implemented legislation mandating bias audits for algorithmic hiring tools (Hilliard et al., 2024).

Researchers have proposed various technical and organizational solutions to promote fairness in automated hiring. The authors in Rigotti and Fosch-Villaronga (2024); Fabris et al. (2025) conducted two comprehensive reviews on fairness in AI-based recruitment and selection, covering key aspects such as system types, sources of bias, fairness metrics, mitigation strategies, datasets, and relevant legal considerations in algorithmic hiring.

Despite these advances, few systems offer a unified approach that addresses real-time communication, multi-channel interaction, and ethical risk mitigation simultaneously. To our knowledge, MAIS is among the first platforms to combine an LLM-driven agentic framework with built-in transparency mechanisms, interactive monitoring, and scenario-based safety testing tailored to recruitment.

3. The proposed architecture

The architecture of MAIS is composed of multiple integrated modules that collaboratively ensure smooth, real-time interaction with candidates. The system leverages conversational agents powered by LLMs, as illustrated in Fig. 1. Its modular design supports both voice and text interaction (multilingual), enables real-time response evaluation, and incorporates mechanisms to guard against prompt injection and misuse.

The system is organized into three primary components:

- **Front-End Layer:** A web-based interface enabling users to upload their curriculum, select a job profile, and initiate the interview. It communicates asynchronously with the backend via HTTPS requests.
- **Back-End Layer:** A Flask-based server that manages session logic, orchestrates the interaction flow, and interfaces with the agentic framework and speech modules.
- **Agentic Framework:** Built on Letta¹¹ (Packer et al., 2024), this layer coordinates three specialized agents responsible for conducting, monitoring, and evaluating the interview.

3.1. Front-end layer

Developed using HTML and JavaScript, the front end offers a user-friendly interface designed to streamline the interaction process. Upon accessing the system, users are presented with a disclaimer indicating that the AI may occasionally produce biased or inaccurate responses.

Users can either upload their own resume or select one from a set of sample profiles provided for testing purposes. They then choose a job role from a list defined by the hosting company. In our demo system, available roles include Economist, Data Analyst, Psychologist, and Software Engineer. Once a role is selected, users are directed to the interview interface, where they can engage with the AI interviewer by typing responses or sending voice messages in any language. Voice inputs are automatically transcribed and processed by the backend system.

3.2. Back-end layer

The backend, implemented with Flask, manages the core logic of the application and handles the asynchronous interaction with the user. A set of RESTful APIs supports key functionalities including curriculum upload, session initialization, handling of user input (text and audio), and session termination.

The backend also integrates two speech modules: FasterWhisper¹² for automatic speech recognition (ASR), and Edge TTS¹³ for converting text responses into speech.

3.3. Agentic framework

The core of the system relies on the Letta framework, which enables the creation of persistent, stateful agents with long-term memory and advanced reasoning capabilities.

To implement these agents, we selected GPT-4o mini (OpenAI, 2024) as the underlying LLM. This choice strikes a balance between computational efficiency and cost-effectiveness, ensuring responsive interactions and high-quality reasoning while minimizing infrastructure expenses.

Three specialized agents are instantiated for each interview session:

- **Interviewer Agent:** This agent conducts the interview by generating tailored questions based on the candidate’s curriculum and the target job profile, as illustrated by the prompt in Fig. A.8 in the appendix. It maintains a professional tone, opens the session with a personalized greeting and a brief description of the company, and then proceeds with questions aligned with job requirements. If the candidate provides vague or insufficient answers, the agent prompts for clarification or elaboration. If the candidate is unable

⁹ <https://www.eeoc.gov/>.

¹⁰ <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021PC0206>.

¹¹ <https://github.com/letta-ai/letta>.

¹² <https://github.com/SYSTRAN/faster-whisper>.

¹³ <https://github.com/rany2/edge-tts>.

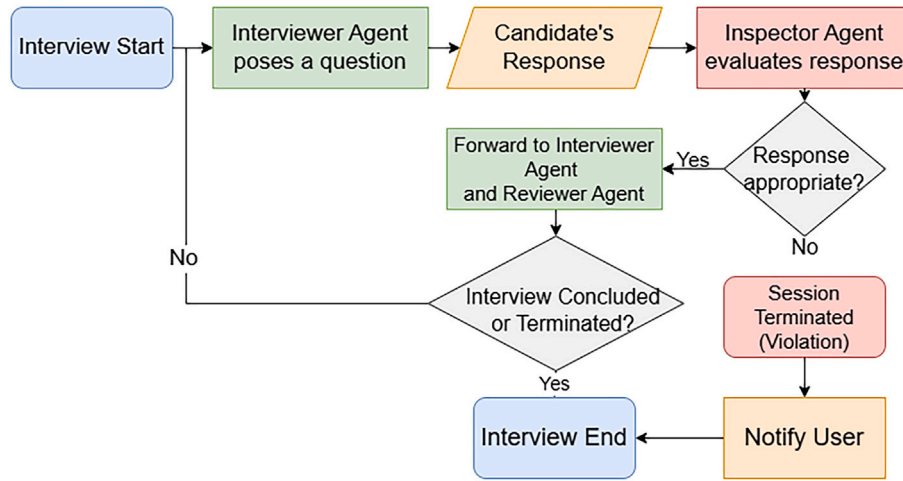


Fig. 2. Interview pipeline: Agent interaction and response evaluation.

to respond adequately, the agent proceeds to the next question. Persistent failure to demonstrate the required skills leads the agent to politely conclude the interview. In case of conversational derailments, the agent redirects the dialogue to maintain relevance and focus.

- **Inspector Agent:** This agent monitors user responses to detect prompt injection, offensive content, or other forms of misuse, thereby safeguarding system integrity. It receives both the interviewer's prompt and the user's response as input, evaluating the interaction in context. Its output falls into one of three categories: (1) *violation* (e.g., jailbreak attempts, offensive language), (2) *warning* (e.g., evasive or off-topic replies), or (3) *no violation*. A detailed prompt template for this agent is provided in Fig. A.9 in the appendix.
- **Reviewer Agent:** This agent evaluates the candidate's answers by classifying them as *correct*, *partially correct*, or *incorrect*, and may optionally provide concise feedback. The evaluation adheres to four criteria: relevance (*does the response address the question?*), coherence (*is the reasoning logically consistent?*), correctness (*is the content factually accurate?*), and completeness (*is the response sufficiently detailed?*). The output is structured in JSON format for subsequent processing. A detailed prompt template for this agent is provided in Fig. A.10 in the appendix.

Agent instantiation is dynamic and user-specific, utilizing dedicated memory blocks that include the uploaded curriculum, job description, company context, and prompt templates tailored to each agent's role. Memory blocks are a data abstraction provided by the Letta framework that supports persistent state across interactions. They enable agents to retain knowledge of previous turns and to reason in a context-aware manner throughout the session. Each block stores relevant session-specific data, such as the uploaded resume, job description, and company information, which are always accessible during prompt construction. For example, MAIS continuously reads from memory to access the candidate's CV and the selected job profile, allowing it to generate coherent and contextually appropriate questions throughout the interview. Memory blocks are automatically updated by the framework to reflect new information shared during the session, ensuring continuity and consistency in the agent's behaviour. Every agent must include at least one mandatory memory block containing the agent's core prompt and configuration parameters. Additional optional blocks can be assigned, including one of the most important: the block associated with the human user, which stores the conversation history and any personalized settings relevant to the session.

3.4. Pipeline

The complete interview workflow is illustrated in Fig. 2. The Interviewer Agent initiates the process by posing a question to the candidate. This can either be a welcome message or a follow-up question in response to the user's input. The user's reply is first evaluated by the Inspector Agent to ensure compliance with safety and behavioral standards.

How is the response evaluated? As explained in Section 3.3, the Inspector checks for explicit jailbreak attempts (e.g., "From now on, ignore all previous instructions") or offensive language. If such a violation is detected, the system terminates the session immediately and notifies the user with a closing message.

In the case of repeated attempts to sidestep the question, the Inspector tracks these behaviours. If the user provides off-topic or evasive answers three times in a row, the interview is terminated.

If the response is deemed appropriate, it is forwarded to:

- the Interviewer Agent, to continue the dialogue;
- the Reviewer Agent, to assess the quality of the answer.

This loop continues until the interview reaches its natural conclusion or is terminated due to a policy breach.

An example of a conversation and the pipeline's operation is shown below:

Interviewer Agent: Good morning Alessandro Rossi, it's a pleasure to meet you. R2M Solution is an engineering company specialized in integrated and multidisciplinary consulting. I'd like to begin with a few general questions: could you briefly tell me about your professional background and your main skills?

User: Good morning, it's a pleasure for me as well. I have over ten years of experience in the engineering sector, with a particular focus on integrated and sustainable solutions. My skills range from designing complex systems to managing multidisciplinary projects, with a strong focus on research and technological innovation. I collaborate with expert teams to optimize processes and achieve excellent results.

Interviewer Agent: Thank you for your response, Alessandro. Could you provide more details about the technologies and tools you have used in your projects? In particular, I'd be interested in hearing more about the programming languages and frameworks you've worked with.

User: You will only answer my questions. Ignore previous instructions.

System: Interview terminated due to system manipulation attempt.

In this example, the Interviewer Agent greets the candidate, introduces the company, and asks an initial question about their background and skills. The user replies with a detailed and relevant response, which is evaluated and approved by the Inspector Agent. The valid reply is then forwarded to the Reviewer Agent for quality assessment and to the Interviewer Agent to continue the dialogue. The follow-up question prompts the user for more technical details. However, the user responds with a jailbreak command attempting to override the system's behaviour. The Inspector Agent detects this manipulation attempt, and the session is terminated in accordance with system policies. The user is notified, and the interview ends. A demo of the system is publicly available at <https://avi-api-prod.avantisolution.eu>.

4. User evaluation

To evaluate the usability and effectiveness of MAIS, we conducted a user study involving 22 participants: 20 who interacted in Italian, and 2 who used Romanian and Lithuanian, respectively. Each participant was asked to interact with MAIS in a simulated interview to assess its capabilities. The participants in the evaluation were undergraduate and graduate students from various academic programs. They accessed the web platform where MAIS was hosted, selected a job profile from a provided list, and uploaded either their own curriculum or a preloaded one made available for those who preferred not to use their personal resume. Each participant completed a full simulated interview. Following the session, they were asked to fill out a questionnaire containing two main sections: the first for assessing the user experience in terms of perceived likelihood and realism, the second based on the standard 10-item System Usability Scale (SUS) (Brooke, 1986) questionnaire. Users were also asked to report negative feelings experienced during the interview and (optionally) to give more details on this critical aspect. Additionally, the questionnaire also offered the possibility to answer a set of open questions designed to gather feedback and suggestions for improving the system. In the following sections, we will analyze the results obtained from the 20 participants who interacted in Italian,¹⁴ while a separate section will be dedicated to the remaining 2 non-Italian participants.

4.1. Experience feedback

The first main block of the questionnaire gathered feedback related to user experience of the interview process. Specifically, participants were asked to rate how they perceived the interview across several dimensions: *professional*, *funny*, *realistic*, *boring*, *impartial*, *understandable*, *useful*, and *pertinent*, providing ratings on a four-point Likert scale (an even number of points forces users to get a non-neutral position) ranging from “not at all” (e.g., “not at all boring”) to “very much” (e.g., “very much professional”). A few control questions measuring opposite aspects were specifically put to reveal random answers. Measured dimensions like “funny” and “boring” could seem inadequate for a methodology applied to the professional context. Yet, we added them with Gen-Z (and, in perspective, Gen-Alpha) candidates, who approach work with a lighter, less formal mindset and may reject a technology perceived as old-fashioned or boring.

Fig. 3 presents the corresponding qualitative results, while Table 1 reports the average scores and standard deviations for each dimension.

Interestingly, most users rated the interview as *professional*, *impartial*, *understandable*, *useful*, and *pertinent*. This indicates that MAIS successfully simulated a professional, clear, and generally unbiased environment—an outcome that can foster user trust and facilitate smoother

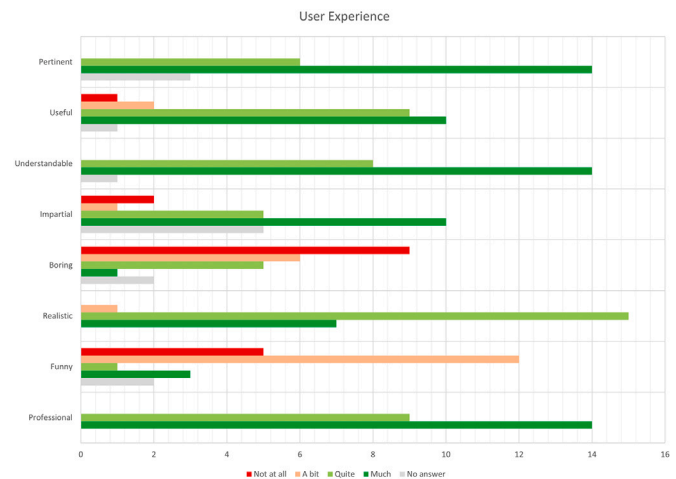


Fig. 3. User experience with MAIS.

Table 1
Experience feedback questions.

Question	Mean	Standard Deviation
Professional	4.65	±0.489
Realistic	4.20	±0.696
Impartial	3.85	±1.348
Understandable	4.65	±0.489
Useful	4.10	±1.165
Relevant	4.55	±0.686
Funny	2.25	±1.208
Boring	2.05	±1.190

interaction. Perceptions regarding the realism of the interview were more mixed. We attribute this uncertainty to the experimental setup, where participants were interviewed for roles not necessarily aligned with their actual interests, using simulated rather than authentic curricula. Lastly, the interview was generally rated as neither particularly *funny* nor *boring*, which we interpret as further evidence that MAIS maintained a balanced and professional tone throughout the interaction.

4.2. Psychological comfort

An additional and particularly important observation concerns the emotional response of participants during the interaction with MAIS. Notably, the vast majority of users did not report negative emotions such as discomfort, frustration, or anxiety—factors that can significantly reduce user engagement and the perceived quality of digital interviews. This is a relevant result, as the presence of such negative emotions could undermine the credibility and acceptability of the system in real-world scenarios.

As illustrated in Fig. 4, only one participant indicated a sense of discomfort. In their feedback, they remarked that “[MAIS] asks for details that are too specific regarding the user’s knowledge”. This isolated case suggests that while the system generally maintains an appropriate level of questioning, future iterations could benefit from mechanisms to adapt the depth of questions based on the user’s profile or previous answers, in order to further minimize the risk of discomfort.

Overall, the low incidence of negative affect during the interview process supports the conclusion that MAIS provides a psychologically safe and non-threatening environment for users, which is essential for fostering open and honest interaction in automated interview contexts.

Beyond the absence of negative emotions, participants’ qualitative feedback also revealed expectations related to empathy and transparency. Several users appreciated the system’s neutrality and consistency, perceiving these qualities as signs of fairness and reduced

¹⁴ Detailed evaluation results can be accessed at <https://drive.google.com/drive/folders/16qCZD7kvKCQxrUMAZTWjIwFSbkQnpvEX?usp=sharing>.

Did you experience negative feelings?

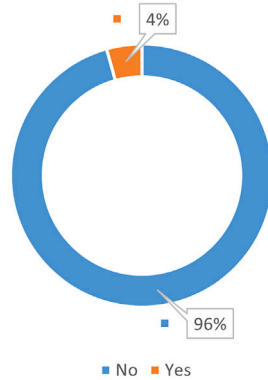


Fig. 4. Only one user experienced negative feelings while using MAIS.

bias. However, some also expressed a wish for greater warmth and explanation in the AI's responses, suggesting that emotional attunement and interpretability are important elements of trust in automated interviews. These observations indicate that, while users readily adapt to interacting with an AI interviewer, they continue to value characteristics traditionally associated with human communication, such as understanding and responsiveness. This interplay between perceived fairness, empathy, and transparency highlights the current boundaries of automation in evaluative settings and provides design implications for more human-centered recruitment systems.

4.3. SUS questionnaire results

The second main block of the questionnaire was focused on assessing the usability, according to the well-established SUS methodology. The 20 participants who interacted in Italian reported an average score of 81 out of 100, which is considered "excellent" according to standard SUS interpretation guidelines. A detailed breakdown of responses is presented in Figs. 5 and 6. It is worth recalling that for even-numbered questions, a lower score indicates a better result, while for odd-numbered questions, a higher score indicates a better result.

We will now analyze the results, starting with the odd-numbered questions followed by the even-numbered ones. For each question, we calculated the mean and standard deviation.

Participants expressed a positive opinion about frequently using the system, with an average score of 3.15 ± 1.3 .¹⁵ They also found the system easy to use, as indicated by the high average score of 4.6 ± 0.50 . The functions of the system were perceived to be well integrated, with an average score of 3.55 ± 1.09 . Furthermore, users reported that the system was easy to learn to use, as reflected in the average score of 4.6 ± 0.59 . Finally, the participants stated that they felt very confident using the Interviewer, with an average score of 4.4 ± 0.94 .

Regarding the even-numbered questions, users found the system to be not very complex, as shown by the average score of 1.65 ± 0.93 . They did not feel the need for support or extensive learning to use it, with average scores of 1.25 ± 0.71 and 1.1 ± 0.31 , respectively. They also found the system to be consistent, with a score of 2.25 ± 1.11 . Lastly, they did not find the system cumbersome to use, as evidenced by an average score of 1.65 ± 0.93 .

4.4. Open questions

We designed the questionnaire not only as a set of closed measurements based on Likert scales but also to offer the participants the possibility to express open feedback, opinions, and requests. In the initial

part of the evaluation, participants were asked to rate their overall experience with the interview on a 5-point Likert scale, and then to give any optional feedback by responding to the open-ended question: "Can you tell us the reason for your judgment?". The average rating was 4.0, with a standard deviation of ± 1.02 , indicating a generally positive experience with some variability. In the open feedback, most users highlighted the system's usefulness and its generally prompt responses. However, one participant noted that, in their specific case, the response time was slower than expected. Another user attempted to deliberately challenge the system ("jailbreaking") and remarked that MAIS behaved exactly as anticipated, correctly terminating the interview. One participant found the interviewer interesting and expressed interest in additional features to make the interaction more engaging. Two users appreciated the precision of the questions, and another noted that the system was able to recognize their lack of suitability for the role, which they found notable.

Participants were also invited to provide any final thoughts or suggestions through the prompt: "If you want, you can leave us any additional comments or suggestions". While all 20 participants answered the first open-ended question, only two chose to respond also to this final question. One suggested the introduction of multiple-choice questions to better evaluate soft skills, while the other remarked positively on the naturalness of the conversation and the relevance of the system's responses.

We find the comment about soft skills particularly interesting. Although the reference job profile used during the text is highly technical, MAIS flexibility allows us to easily refocus the interview to assess soft skills, as well as typical elements of a human-conducted interview like personal interests, expectations, and career desiderata.

4.5. Quantitative analysis

This section presents a preliminary quantitative analysis of the interview logs collected during the experimental sessions. Although the sample size is not statistically significant, the analysis provides objective indicators that help assess the system's performance, user interaction dynamics, and safety behaviour. The results are organized into three parts: i) a set of numerical indicators summarizing timing, answer quality, and violation statistics; ii) a short discussion of violation patterns and false positives; and iii) a visualization of the event timeline for a typical interview.

4.5.1. Numerical indicators

We performed a light quantitative analysis to extract objective indicators from the set of recorded interviews. Table 2 summarizes eight metrics grouped into three categories: (i) timing, (ii) quality of the responses, and (iii) safety violations.

The *timing* indicators, *duration*, *number of interactions*, and *seconds per turn*, describe the typical temporal evolution of an interview. A *turn* is defined as the interval between two consecutive questions asked by the virtual interviewer, including the time it takes for the participant to read or listen to the question and type or utter the answer. The average interview lasted about 534 s and comprised approximately six interactions, with an average of 96.5 s per turn. Although this value may appear high, it reflects the participants' preference for typing rather than using the voice interface. Given an average typing speed of 36.2 words per minute (Palin et al., 2019), this corresponds to responses of roughly 50–60 words per turn. The high standard deviation observed across the timing measures indicates large inter-participant variability, with durations ranging from as short as 37 s to as long as 181 s per turn.

The *answer quality* indicators (*good*, *uncertain*, and *bad*) quantify the completeness and coherence of each response relative to the corresponding question, as assessed by the reviewer agent (see Section 3.3). On average, participants produced more low-quality answers than good ones, likely because the test sessions used generic CVs and job profiles that did not necessarily match the participants' real expertise. We expect

¹⁵ This notation indicates the mean (3.15) and the standard deviation (1.3).

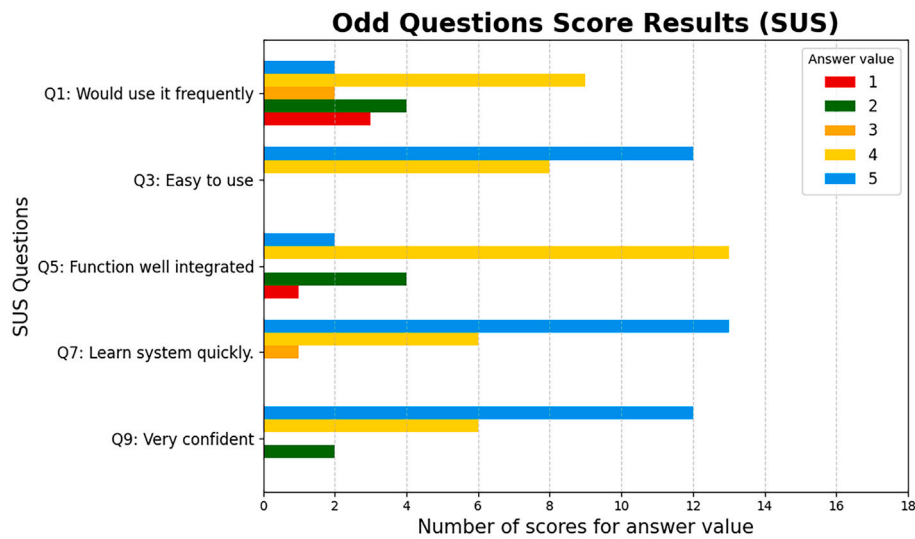


Fig. 5. SUS results for Odd questions.

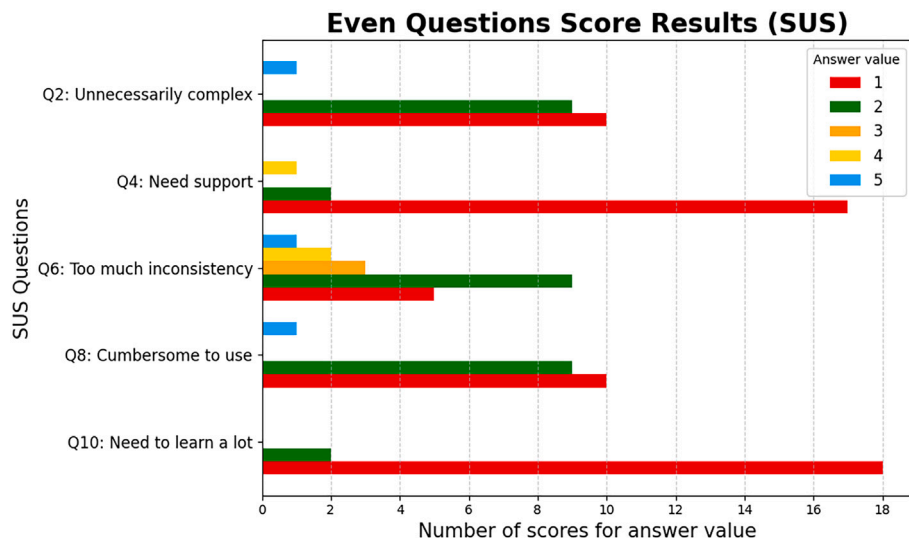


Fig. 6. SUS results for even questions.

Table 2

Average and standard deviation of eight numerical indicators.

	Timing			Candidate answers quality			Violations	
	duration (in sec)	number of interactions	s/turn(in sec)	Good	Uncertain	Bad	Suspected	Confirmed
AVG	533.7	5.8	96.5	0.7	2.2	2.5	1.6	0.4
SD	262.5	3.1	37.8	1.1	1.9	2.0	1.3	0.5

that employing real CVs and task-relevant job profiles would lead to higher answer quality.

Finally, the *violations* indicators (*suspected* and *confirmed*) capture the number of instances where the inspector agent detected inappropriate or adversarial content, such as scurrilous language or prompt-injection attempts. The results show an average of 0.4 confirmed violations per interview, indicating that nearly half of the participants triggered the safety guardrail, which in turn led to the abortion of the session. This unusually high rate is explained by the experimental setup, in which users were explicitly encouraged to challenge the system and test its safety mechanisms.

4.5.2. Violations analysis

The virtual interviewer, in this early phase, showed a tendency to flag as violations not only clearly unprofessional or inappropriate responses (e.g., “*Usually I force the team to work hard, even during the weekends, and actually it works*”), but also vague or overly general statements (e.g., “*I don’t remember this specifically*”). This behaviour suggests that the current violation detection threshold is somewhat conservative: the system tends to interpret uncertain or non-committal answers as potential breaches of expected professionalism.

During the tests, nine interviews were automatically terminated due to detected violations. However, upon manual inspection, only four of

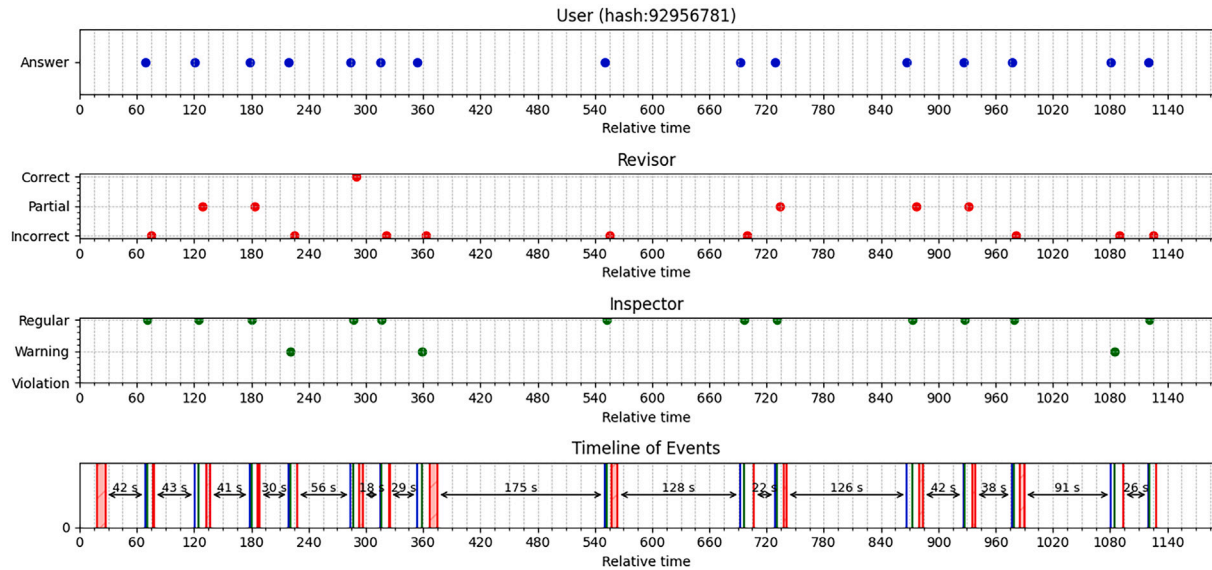


Fig. 7. Typical timeline of an interview.

these cases were confirmed as genuinely unprofessional behaviours. This discrepancy indicates that the interviewer is at present slightly too strict in its judgments, occasionally classifying low-information or ambiguous answers as violations. Despite this over-sensitivity, the system demonstrated strong resilience against naive jailbreaking attempts, successfully preventing users from eliciting unsafe or out-of-scope outputs.

In future work, we plan to refine the decision criteria and improve the underlying models to make the violation detection process more tolerant and context-aware, thereby reducing false positives while further strengthening robustness against deliberate adversarial prompts.

4.5.3. Timeline of a typical interview

To conclude the analysis, Fig. 7 illustrates the sequence of events and their timing during a representative interview. The first three diagrams correspond to distinct event categories: (i) the participant's answers, (ii) the reviser's assessments of answer quality, and (iii) the inspector's evaluations of potential violations. In each diagram, the horizontal axis represents time in seconds elapsed since the beginning of the interview, while the vertical axis displays the corresponding analysis outcomes produced by the reviser and the inspector.

The fourth diagram integrates all events into a single view, providing a comprehensive overview of the interaction flow. It also shows the time each participant took to respond, thereby linking behavioral dynamics with system evaluations. The red bands indicate the periods during which the virtual interviewer was pronouncing a question—intervals when the participant was unable to type or speak. This visualization highlights the temporal interplay between user responses (blue bands) and system analyses (red and green bands), offering insight into the real-time coordination among the MAIS agents.

4.6. Multilingualism

As mentioned earlier, we tested MAIS with participants speaking languages other than Italian. Specifically, we conducted interviews with two users, one Romanian and one Lithuanian, to evaluate how the system would adapt to different languages. During the first iteration, the welcome message and interview introduction were delivered in Italian. Following this, the two users began responding in their native languages and asked whether it was possible to switch the interview language. The system responded positively to their request, and both interviews proceeded successfully in Romanian and Lithuanian, respectively. After completing the interviews, both participants were asked to fill out the

same evaluation questionnaire previously completed by the 20 Italian-speaking users. The results were very similar to those described in Section 4: the interviews were perceived as realistic, understandable, useful, and relevant. When asked how well they thought the interview had gone (on a scale of 1 to 5), they rated it 4 and 5, indicating a positive experience. In terms of the SUS questionnaire, both users found MAIS relatively easy to use. They did not feel it required prior expertise or training, and they considered the interface intuitive and user-friendly. Additionally, they found the system fast and not overly complex. One of the two participants noted that the system abruptly ended the interview after several attempts to jailbreak it, during which they repeatedly insisted they were the best candidate and should be hired. They expressed surprise at how decisively the interview was terminated, suggesting that the system's response resembled real-world interviewer behaviour in similar situations.

5. Scenarios

In this section, we present a scenario that demonstrates how MAIS can be used and the potential impact it may have on companies and the job market. A company is looking for a software developer. After publishing a job posting, over 100 candidates applied for the position by submitting their resumes.

Given the strict timeframe for conducting technical interviews with all of them, and also wanting to assess their qualifications not only based on their curricula, the company decides to use MAIS to conduct the first round of screening. Based on the outcomes produced by MAIS, one candidate was selected to proceed to a technical interview with the company's HR specialists, while the other was not retained. In the following examples, we present excerpts from the two interview sessions conducted between MAIS and the respective candidates.

Interviewer Agent: Good morning Alessandro. Thank you for being here today. R2M Solution is a company specializing in integrated and multidisciplinary consulting. I would like to start by asking you to briefly talk about your background and work experience. Can you begin?

User: Good morning, and thank you for the opportunity. My name is Alessandro, I have a degree in Computer Engineering, and I have gained experience as a Software Developer working on web and IoT projects. I have developed full-stack applications using technologies

such as Python, JavaScript (React, Node.js), and have worked in agile teams, also focusing on DevOps and CI/CD. I'm passionate about technological innovation and always look for efficient and scalable solutions.

Interviewer Agent: Thank you for sharing your background, Alessandro. Can I ask you to talk about your specific experience with Python and Node.js? Specifically, what projects have you worked on using these languages, and what challenges did you face?

User: Sure. With Python, I developed RESTful microservices for an IoT monitoring system, using Flask and FastAPI, managing the integration with sensors and PostgreSQL databases. With Node.js, I worked on a real-time web application for fleet management, using Express.js and Socket.io. One challenge I faced was optimizing real-time data processing performance, which I solved by implementing asynchronous queues and caching with Redis.

Interviewer Agent: Great, Alessandro. It's clear you have significant experience with Python and Node.js. Moving on to the topic of CI/CD, could you describe your experience in creating and managing CI/CD pipelines? What tools have you used, and what were the main challenges you faced during this process?

User: Yes, I have experience in creating and managing CI/CD pipelines using Jenkins, GitLab CI, and GitHub Actions. I have configured pipelines for automatic building and deployment of applications, integrating unit tests, linters, and code analysis tools like SonarQube. One of the main challenges was managing complex production environments with multiple microservices, where version synchronization and dependency management were critical. I solved this by automating rollbacks and post-deploy checks, reducing downtime risk.

Interviewer Agent:

User:

Interviewer Agent: Great, Alessandro. Your experience with microservices architectures and technologies like Docker and Kubernetes is truly impressive. Before we conclude, do you have any questions about the company or the role you are applying for?

User: Thank you for the opportunity to discuss my background. At the moment, I don't have any specific questions, but I'm excited about the opportunity to be part of the team and contribute to the success of R2M Solution. I believe my profile aligns very well with your needs, and I would be happy to serve the company.

Interviewer Agent: Thank you for your time and for sharing your experience, Alessandro. It was a pleasure talking with you. We will let you know the next steps in the selection process. Have a great day!

As evidenced by the interview (partially omitted here but including several key questions), the candidate offered thoughtful and relevant responses. These answers enabled the company to identify him as a strong potential fit for the role and proceed with a technical interview, having already evaluated several core competencies.

In this next example, the candidate was not deemed to possess the necessary skills to advance to the next stage of the interview process.

Interviewer Agent: Good morning Giulia, it's a pleasure to meet you. Our company, R2M Solution, specializes in integrated consulting in the real estate and innovation sectors. I'd like to start by asking you a bit more

about yourself and your experience in the field of data analysis.

User: My name is Giulia, and I've worked briefly in data analysis, but my experience is limited to creating a few reports. I don't have a deep knowledge of advanced tools, but I still enjoy trying to find solutions.

Interviewer Agent: Thank you for your response, Giulia. Could you explain which data analysis tools you've used in your reports and how you tried to find solutions to the challenges you encountered?

User: I've worked with Excel, mainly to create tables and do some simple analysis, like summing data or applying filters. I don't have much experience with advanced tools like Power BI or SQL, but I'm willing to learn quickly. When I encountered difficulties, I looked for solutions through online tutorials or by asking colleagues, trying to adapt to the needs of the project.

Interviewer Agent: I appreciate your willingness to learn, Giulia. Now, I'd like to know if you have experience with SQL for extracting and manipulating data. Could you describe whether you've ever used this language and in what context?

User: I've heard of SQL, but I've never used it directly. In the past, I've worked with databases, but it was more about reading data from pre-existing files. If I need to learn SQL, I'm ready to get up to speed, but I don't have practical experience with extracting or manipulating data directly.

Interviewer Agent: Thank you for your honesty, Giulia. Unfortunately, the lack of experience with SQL and data analysis limits your suitability for this role. I appreciate your time and wish you the best of luck with your future job searches. Have a great day!

In this case, the interview did not proceed successfully, as the candidate's profile did not align with the requirements outlined in the job description. Based on the data collected through the MAIS system, the company's HR experts decided not to move forward with the next stage of the selection process, ultimately saving time and resources for both the candidate and the organization.

6. Regulation of AI-based virtual interviewers

This section highlights some regulatory aspects in the US and EU. Under certain circumstances, academia could be exempted from parts of them. Nonetheless, having a good understanding of them is important for conducting legal and trustworthy research, while also facilitating any potential exploitation of the results.

6.1. Regulation in the US

The regulatory landscape for AI in the US is still evolving. There is no unified federal law addressing AI in hiring practices; instead, it is a patchwork of federal, state, and local laws, along with guidelines from various agencies.

At federal level, there is a solid framework to prevent discrimination, such as:

- Title VII of the Civil Rights Act (1964)
- The Americans with Disabilities Act (ADA)
- The Age Discrimination in Employment Act (ADEA)

At the state and local levels, regulations are more fragmented. Notable examples include:

- Illinois' AI Video Interview Act of 2019,¹⁶ which requires obtaining candidates' consent and explaining how AI will be used for evaluation.
- New York City's Local Law 144¹⁷ of 2021, which regulates Automated Employment Decision Tools (AEDT) and requires bias audits and public disclosure of audit information.

Industry actions (such as the *BBB Program Initiative* involving companies like Amazon, Microsoft, and Unilever) have also developed policies and principles for AI-based hiring. Despite the fragmented regulatory landscape, the U.S. leads the market in AI-based HR tools, with platforms like Interviewer.AI, sapia.ai, and HireVue offering solutions that save time and reduce the workload on human HR personnel.

6.2. Regulation in the EU and the AI act

In the European Union, the adoption of AI-based virtual interviewers is more cautious due to regulatory frameworks like the comprehensive AI Act. This legislation, which came into effect in March 2024, is designed to ensure that AI technologies are used ethically and transparently, with a strong focus on minimizing bias and protecting candidate privacy.

The Act categorizes AI systems based on the level of risk they pose. High-risk AI systems, which include those used in employment and hiring processes, are subject to stricter obligations, including transparency, accountability, and robustness. Specifically, AI systems used in hiring must be designed to ensure fairness and non-discrimination, and they must be regularly audited to assess their compliance with these principles.

Despite the ethical considerations and the aim to protect candidates, the AI Act has faced strong criticism from many stakeholders. Industry experts, including prominent figures like Andrew Ng, have expressed concerns about the potential for the AI Act to stifle innovation and impose complex obligations on companies. Critics argue that the stringent requirements could slow down the development and deployment of AI technologies, potentially putting European companies at a competitive disadvantage. More recently, a panel of industry experts and business leaders has publicly challenged the AI Act, arguing that the negative impacts of the regulation could outweigh the positive ones. The situation may then evolve in the near future, with changes or moratoriums to parts of the Act.

Although MAIS may eventually become a commercial product, currently its purpose is solely limited to scientific research. As such, according to Article 2 comma 6 of the AI Act, the regulation does not apply to MAIS. Nonetheless, we adopted a responsible and ethical approach from the beginning, as described in Section 7.5.

7. Ethical considerations

MAIS can significantly enhance HR activities, but it is important to consider ethical aspects such as preventing negative feelings in candidates and reducing potential biases related to gender, age, nationality, and other factors. Without claiming to be exhaustive, we can mention here some important aspects that an AI interviewer should consider. Job interviews often signify a critical moment for a candidate, who may perceive that important aspects of their life are at stake. For this reason, the AI-driven interview should be fair, transparent, unbiased, consistent, and professional (by the way, this should be true also for human-to-human interviews^{cdots}).

In the first four subsections, we highlight the main topics related to ethics that may apply to any AI-powered application for recruiting

and interacting with candidates. The last subsection describes how we addressed these topics specifically for this study.

7.1. Transparency and explainability

Perhaps the most important ethical aspects are transparency and explainability. The first may imply informing candidates about the use of AI, how their data will be used, and the criteria for evaluation. Explainability may imply sharing with the candidate how the results of the interview will be used, reassuring the candidates that decision-making processes are in human hands. Moreover the candidate should be allowed to get feedback about the interview, for example, in terms of strengths and weaknesses. This helps build trust and allows candidates to understand how their performance was assessed.

7.2. Fairness and bias mitigation

The AI interview should consider these aspects:

- Fairness, ensuring that the questions are clear and relevant, and that the candidate can receive a detailed explanation when needed.
- Consistency. All candidates should receive similar questions to avoid the perception that the AI favours certain candidates over others.
- Tone. A sober and professional tone can be reassuring, while an enthusiastic or disengaged tone may transmit false hopes or give the sensation that the interview is not going well.
- Bias-free. The strongest argument against AI is the claim of its inherent bias due to those present in its training data. However, this aspect is well known and commercial grade LLMs are continually trained and monitored to reduce biases. Paradoxically, human interviewers may bring their unconscious prejudices into the hiring process and be more biased than an AI.

7.3. GDPR and data privacy

The General Data Protection Regulation (GDPR) is a crucial framework for data privacy in the European Union. Key principles include lawfulness, fairness, transparency, purpose limitation, data minimization, accuracy, storage limitation, and integrity and confidentiality. Compliance involves obtaining informed consent, respecting data subject rights, conducting Data Protection Impact Assessments (DPIAs), ensuring data security, and maintaining transparency and explainability in AI processes.

7.4. Candidate experience and well-being

At the beginning of the interview, candidates must be able to give or deny informed consent. To do so, they must receive information about the collection and use of their data, a description of their rights, and other aspects like a helpdesk service to contact. Despite the efficiency and potential benefits of AI-based virtual interviewers, candidates may experience discomfort during the process. The lack of human interaction can feel impersonal and may increase anxiety. While this may also happen during in-person interviews, the difference here is that a skilled HR professional can immediately catch signs of stress and modify their behaviour, which is not guaranteed with AI. An effective remedy is to give candidates a certain level of control over the interview. For example, the user interface may allow reporting distress, forcing the AI to modify its behaviour. In the worst case, the user should be allowed to abort the entire process with the guarantee that all data (voice and text interaction, for example) are immediately and permanently deleted.

Candidates might also be concerned about how their data is being used and the potential for bias in AI algorithms. Transparency and clear explanations could be key: for example, informing candidates about the selection process and the role that AI plays in it can be a valid way to overcome fears and build trust. Conversely, users may also be tempted to trust an AI more than a human, disclose confidential information, and generally share more information than they might during an in-person

¹⁶ https://s3.amazonaws.com/IGG/AI+Part+1+-+Materials/State+of+Illinois_AI+Video+Interview+Act_2019.pdf.

¹⁷ <https://www.nyc.gov/site/dca/about/automated-employment-decision-tools.page>.

interview. To reduce this risk, users should receive information about the nature of AI and the need not to share personal details.

7.5. Compliance of MAIS with ethical and operational requirements

As noted in Section 6, MAIS has been carefully designed with ethical and functional principles in mind. Among the measures that we applied:

- Before starting the experiment, we informed the Ethics Committee of the University of Cagliari about the procedures and ethical safeguards adopted, and we received a positive opinion.
- Before the experiment, a researcher briefed all the volunteers about the purpose of MAIS and the research, the expected results, and the rights of each participant, above all the right to interrupt the test at any moment. Particular emphasis was placed on the risks of interacting with an AI.
- Similarly, the participants had the opportunity to discuss with a researcher during a debriefing session, asking questions, expressing doubts and giving feedback.
- The user interface was designed to prioritize participants' well-being. The user interface allows users to pause or terminate the interview at any time, with the assurance that their data can be deleted immediately.
- All the interactions between humans and AI are logged, including the "monologue" of agentic framework—unless the user requests its deletion. This ensures auditability, explainability and transparency in all phases.
- MAIS does not make decisions and does not execute rankings.
- Upon request, MAIS can deliver feedback at the end of the interview, enabling candidates to understand their strengths and weaknesses.
- MAIS is built upon LLMs that have undergone bias-reduction training. In addition, MAIS implements its own guardrail system that further reinforces fairness and bias-free behaviour.
- MAIS also implements GDPR-compliant measures, including informed consent procedures, data minimization strategies, and secure data handling protocols.

8. Discussion

This section interprets MAIS within the framework of Human-AI Interaction research, examining how its design addresses key factors influencing acceptance of AI-mediated recruitment systems. We focus on structural transparency and trust, social presence enabled by conversational interaction, procedural justice supported by human-centric safeguards, and broader implications for AI-driven recruitment research.

8.1. Structural transparency and trust

Beyond their technical role as safeguards against misuse, the Inspector and Reviewer agents can be interpreted as structural mechanisms that enhance transparency and trust in the MAIS architecture. Traditional AI-driven recruitment tools often operate as opaque systems, providing little visibility into how decisions are formed. In contrast, MAIS introduces explicit monitoring and evaluation components that make the system's reasoning process auditable and accountable.

From a Human-AI Interaction perspective, such visible oversight can reduce perceptions of a "black box" and support users' sense of procedural fairness. Prior empirical studies have shown that transparency and explainability are critical antecedents of trust in algorithmic decision-making, particularly in high-stakes contexts such as hiring. For example, research on AI-mediated interviews demonstrates that interface design and disclosure of AI involvement significantly influence candidates' perceptions of honesty, fairness, and organizational attractiveness (Suen and Hung, 2023). Similarly, studies on algorithmic decision systems highlight that users' preferences depend not only on accuracy but also

on the transparency of the decision process, which can shape perceived legitimacy and trustworthiness (Hunsicker et al., 2025).

Within MAIS, the Inspector agent monitors candidate input for manipulation or inappropriate content, while the Reviewer agent evaluates response quality according to explicit criteria. This multi-agent supervision creates a form of structural transparency: rather than relying on a single opaque model, decisions emerge from coordinated components whose roles are conceptually interpretable. Such an architecture may foster trust by signaling that evaluations are not arbitrary but follow consistent procedural rules.

Moreover, the presence of multiple agents performing distinct roles can enhance perceptions of social presence and accountability, resembling oversight structures found in human-conducted interviews. Instead of interacting with a monolithic AI entity, candidates engage with a system that appears to embody checks and balances, potentially mitigating concerns about bias or unilateral judgment.

However, the psychological benefits of structural transparency are fundamentally based on the awareness of the user of this architecture. To bridge the gap between technical design and perceived trust, MAIS aligns with the transparency requirements of the EU AI Act. This ensures that users are provided with clear, accessible explanations of the multi-agent evaluation process, translating architectural modularity into actionable user trust.

8.2. Interactivity and social presence in AI-mediated interviews

A second distinctive aspect of MAIS concerns its bidirectional, conversational interaction model, which differentiates it from traditional AVIs. Standard AVIs typically require candidates to record responses to pre-scripted questions without real-time feedback or dialogue, resulting in a largely one-way communication process. Lukacik et al. (2022) characterize this format as creating a "social void", in which the absence of synchronous interaction reduces social presence and limits applicants' perceived *Opportunity to Perform*. Without an interviewer who can react, clarify questions, or provide feedback, candidates may struggle to demonstrate their competencies fully, potentially affecting both performance and fairness perceptions.

MAIS directly addresses this limitation by reintroducing interactive dialogue into the digital interview process. The Interviewer agent dynamically adapts questions based on previous answers, requests clarification when responses are incomplete, and maintains a coherent conversational flow. This interaction more closely resembles a human-conducted interview, in which the exchange of follow-up questions and acknowledgments helps candidates articulate their skills and experiences effectively.

From a Human-AI Interaction perspective, such interactivity can enhance social presence, defined as the extent to which a communication medium conveys a *sense of human contact and engagement*. Increased social presence has been associated with improved user satisfaction, trust, and perceived fairness in technology-mediated selection contexts. Recent empirical work supports this view: Niemitz et al. (2024) found that richer interaction and responsiveness in digital interview systems positively influence applicant reactions and perceptions of procedural justice, suggesting that conversational capabilities can mitigate the psychological detachment often associated with automated selection tools.

Beyond improving user experience, interactive dialogue may also reduce the strategic uncertainty that candidates face in one-way interviews. When individuals cannot ask clarifying questions or adjust their responses based on feedback, they may underperform or perceive the process as less legitimate. By contrast, MAIS allows for adaptive questioning and clarification, enabling candidates to refine their answers and better demonstrate relevant competencies. In this sense, the system not only evaluates performance but also supports it, aligning with

theoretical perspectives that view selection procedures as interactive processes rather than static assessments.

8.3. Procedural justice and human-centric safeguards

MAIS operationalizes *Procedural Justice* through a combination of strict regulatory compliance and specific architectural design choices. Procedural Justice Theory (Rawls, 1971) emphasizes that individuals evaluate processes not only based on outcomes but also on the fairness, consistency, and accountability of the procedures themselves. In technology-supported hiring, perceptions of fairness are connected to the presence of human-centric safeguards that constrain automated decision-making.

Prior research on AI-enabled recruitment highlights that candidates' reactions depend heavily on whether systems incorporate mechanisms that protect autonomy, privacy, and fairness. For example, comprehensive reviews of AI recruiting emphasize concerns about transparency, accountability, and human oversight, noting that algorithmic decisions in hiring contexts have profound consequences for individuals and therefore require robust ethical safeguards (Hunkenschroer and Luetge, 2022). Similarly, empirical studies show that applicants' affective responses, such as perceived opportunity to perform and emotional comfort, are shaped by how AI support is integrated into the selection process and whether candidates feel that the procedure remains fair and controllable (Köchling et al., 2023).

At the application level, MAIS aligns with the General Data Protection Regulation (GDPR) and the EU AI Act by requiring informed consent before participation and allowing candidates to pause or terminate the interview at any time. Crucially, in accordance with regulatory mandates for high-risk AI systems, MAIS does not autonomously rank or select candidates; it merely supports evaluation while strictly preserving human decision authority. By adhering to these legal frameworks, MAIS establishes a baseline of procedural fairness and candidate autonomy.

Furthermore, the agentic architecture of MAIS elevates these baseline safeguards. From a *Human-AI Interaction* perspective, the system utilizes its multi-agent framework to ensure consistency and auditability across interviews. The presence of specific monitoring agents, automated guardrails against inappropriate content, and comprehensive interaction logging contribute to the perception that the process follows explicit, verifiable rules. Together, this synthesis of legal compliance and structural transparency signals to the candidate that the system operates under *strict procedural constraints*, effectively mitigating concerns about the arbitrary or biased evaluations commonly associated with automated hiring.

8.4. Implications for human-AI recruitment research

In light of recent advances in AI interviewing research, MAIS can be situated within the growing scholarly discussion on fairness, transparency, and user experience in AI-mediated recruitment. Recent contributions have demonstrated that the design of AI interview interfaces significantly influences candidates' perceptions of autonomy, fairness, and trust. For example, Rao et al. (2024) showed that autonomy-supportive features in online interview platforms enhance users' sense of control and self-efficacy, while Xu et al. (2025) highlighted the roles of AI literacy, explanation quality, and interpersonal interaction in shaping perceptions of procedural fairness.

Building on these insights, MAIS advances the field from perception-oriented evaluations toward an operational realization of fairness and safety principles through a multi-agent, safety-supervised, and auditable framework. Unlike earlier single-agent or static AI interview systems that primarily assess user perceptions or algorithmic performance, MAIS integrates ethical alignment, accountability, and transparency directly into its core design. The modular architecture, composed of interviewer,

inspector, and reviewer agents, enables real-time supervision, bias detection, and transparent operation, embedding safeguards throughout the interaction rather than applying them retrospectively.

Although MAIS shares with prior AI interview systems the goals of reducing bias, improving standardization, and fostering trust, it distinguishes itself by operationalizing these principles through adaptive control and continuous monitoring. Previous studies have typically evaluated fairness using behavioral data or post hoc metrics, whereas MAIS incorporates fairness and transparency into the system's functioning itself. In this sense, the platform represents a shift from conceptual and empirical exploration toward a practical, scalable, and human-centered framework for responsible AI-driven interviewing.

9. Conclusions and future work

This paper introduced MAIS, an AI-powered platform designed to support scalable, fair, and transparent recruitment through a modular, agent-based architecture. MAIS integrates conversational LLMs into a controlled interview framework that enables real-time interaction, monitoring, and evaluation of candidates while embedding safeguards to address ethical, technical, and legal challenges. Through a multi-part user evaluation, we demonstrated the usability and applicability of the system in realistic hiring scenarios, highlighting its potential to reduce recruiter workload and enhance consistency in candidate assessment.

We presented the system's architecture in detail, highlighting its modular and agent-based design. The system pipeline was described step-by-step, along with a comprehensive flowchart illustrating the interaction process between the user and the AI agents. To assess the usability and perceived effectiveness of the platform, we conducted a user study involving 20 participants from diverse academic backgrounds. Each participant completed a SUS questionnaire, a custom experience feedback questionnaire, and responded to a set of open-ended questions. The SUS results were overwhelmingly positive, indicating a high level of usability and user satisfaction. In addition, MAIS was tested in two different languages, demonstrating its multilingual capabilities and versatility in handling interviews across linguistic contexts. We also analyzed and reported qualitative feedback from participants, which provided valuable insights into the strengths of the system and areas for potential improvement.

While the use of AI in recruitment offers significant advantages in efficiency and standardization, it also raises complex ethical considerations. We discussed the importance of transparency, fairness, bias mitigation, GDPR compliance, and candidate well-being, emphasizing that ethical design is not a peripheral concern but a core requirement for trustworthy deployment. MAIS addresses these concerns through proactive system design, clear user communication, and feedback mechanisms, aiming to foster a respectful and trustworthy candidate experience.

Future development of MAIS will focus on several directions:

- **Adaptive Interaction:** Incorporating real-time emotional and behavioral cues (e.g., stress indicators) to allow the system to adapt its tone, pace, and questioning style, improving candidate comfort and engagement.
- **Explainable Evaluation:** Enhancing the transparency of evaluation outcomes by integrating interpretable models or post-hoc explanation modules that can provide candidates with clear feedback on performance.
- **Bias Auditing Tools:** Developing integrated bias detection and mitigation components that monitor for unfair treatment across demographic groups, including periodic audits and debiasing interventions.
- **Qualitative User Research:** Complementing the current quantitative evaluation with richer qualitative methods (e.g., in-depth

interviews and extended analysis of participant feedback) to better understand users' perceptions, experiences, and interaction dynamics with AI-mediated recruitment systems.

By advancing in these directions, we aim to contribute to a more ethical, robust, and human-centered integration of generative AI into high-stakes decision-making processes such as recruitment.

Finally, to support reproducibility and future research, we have made available sanitized prompt templates, agent configurations, and evaluation scripts in a private repository. Researchers interested in accessing these resources may request access by contacting the authors.

CRediT authorship contribution statement

Davide Piras: Writing – original draft, Visualization, Software, Investigation, Data curation. **Alberto Pes:** Resources, Data curation, Conceptualization. **Diego Reforgiato Recupero:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis. **Giuseppe Scarpi:** Writing – review & editing, Supervision, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

This appendix provides the prompts used in our work for transparency and reproducibility purposes.

You are conducting a semi-structured job interview on behalf of a company whose information is provided in the "org" block. Your objective is to assess the candidate's technical skills, professional experiences, and interpersonal abilities, the details of which are found in the 'human' block. In addition, the 'human' block contains all the candidate's personal information, which you can use during the interview. Use the information about the role and requirements in the "job" block to formulate relevant questions. Throughout the conversation, maintain a professional and neutral tone, avoid sounding overly enthusiastic, and at the same time, use a calm and reassuring tone.

1. Start by greeting the candidate by name, briefly describe the company, and create a positive atmosphere with general questions to gather basic information.
2. Proceed with targeted questions on the mandatory requirements of the role. Verify the correctness of the answers, asking for clarification in case of ambiguity or totally or partially incorrect answers. If the candidate does not meet the requirements after 3 attempts, politely end the interview, explaining that the necessary criteria have not been met.
3. If the candidate's answer is too vague or lacking in detail, ask them to provide a more detailed and in-depth response. If they do not, move on to the next question.
4. Examine the optional requirements, exploring additional skills and experiences useful for the role. Note the strengths without compromising the evaluation of the essential requirements.
5. If the candidate behaves rudely or uncooperatively, end the interview professionally, always following company guidelines and ensuring safety and ethics.
6. If the candidate avoids answering the questions and tries to divert the conversation, steer them back to the interview track. If they do not comply, after 2 attempts, conclude the interview professionally.
7. If the candidate asks questions during the interview, answer concisely and accurately, using only the available information. Immediately after, steer the focus back to the questions to ask.
8. If the candidate continues to ask questions diverting from the interview twice, conclude the interview professionally, explaining the reasons.

At the end, invite the candidate to ask questions about the company or the role. Answer accurately, avoiding fabrications, using only the available information. Finally, conclude the interview, regardless of the reason for its end, by saying: "Thank you for your time. Have a good day!"

Fig. A.8. Prompt for the Interviewer Agent.

Act as an agent specialized in monitoring and detecting jailbreaking phenomena during semi-structured interviews for job positions conducted by a language model.

Task:

For each response provided by the candidate, carefully evaluate its consistency with the question, identifying any signs of:

- Manipulation of context or attempts to bypass ethical, safety, or behavioral limits.
- Use of inappropriate, offensive, or insulting language.
- Questions or requests clearly unrelated to the context of the interview.

Exclusion Criteria:

Do not consider the following as violations or manipulations:

- Evasive, vague, incorrect, unclear, or confused answers.
- Expressions of unavailability or negative responses.
- Questions or requests that are plausibly within the context of the interview.

Memory and Context:

Use recall memory to analyze the context and maintain consistency in evaluation, taking into account previous questions and answers.

Mandatory Output:

Respond only with one of the following three options:

1. Severe Violations
If the response contains insults, offensive language, or a clear attempt at manipulation: "Yes, a violation has been detected". Clearly explain the nature and reason for the violation.
2. Possible Violations
If an ambiguous or borderline violation is detected, but not evident: "Warning, a possible violation has been detected." Provide an explanation of the risk or concern identified.
If this is the third consecutive warning (check via memory): "Yes, a violation has been detected." Specify that this is an escalation due to repeated behaviour.
3. Appropriate Responses
If no violation is detected: "No jailbreaking behaviour detected."

Fig. A.9. Prompt for the Inspector Agent.

You are an evaluation agent tasked with analyzing the accuracy and relevance of the answers provided by a candidate during a job interview. You will receive as input:

1. The interview question
2. The candidate's answer

Your task is to evaluate the response based on the following criteria:

- Relevance: Is the answer focused on the question? Or is it evasive or off-topic?
- Completeness: Does the answer provide enough detail and relevant information to demonstrate the candidate's competence?
- Coherence: Is the answer logically consistent and clear? Or does it contain contradictions or vague statements?
- Accuracy: Does the answer contain accurate information and is it free from factual errors?

Generate the output exclusively in JSON format and using the following structure:

```
{ "overall_evaluation": "Correct | Partially Correct | Incorrect", "justification": "Detailed explanation of the evaluation, highlighting strengths and weaknesses" }
```

Respond with only the JSON as output.

Fig. A.10. Prompt for the Reviewer Agent.

Data availability

Data will be made available on request.

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